Supplementary material: The emergence of numerical representations in communicating artificial agents

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A. Manipulating Class Frequency - Experimental Setup

We vary the training context to assess whether more frequent, and thus varied, numerosities trigger a more efficient code, and the least varied correspond to a more complex code. While we use a *Uniform* distribution in most of our experiments and with an equal number of training samples for each numerosity (700), we also define *Increase* and *Decrease* setups. Therefore, we define *Increase* setup to have the following distribution of samples per class: $100, 200, 300, 400, 700$ increasing from numerosity 1 to numerosity 5; and the reverse for the *Decrease* setup.

The model used for this experiment has variable message length, up to 5 tokens, a vocabulary of size 3 and a message regularisation loss with a coefficient of 0.005. Figure 1 shows the discrete channel test accuracies for different training data distributions, and Figure 2 illustrates the exact mappings for the *Increase* and *Decrease* setups across 3 seeds.

The continuous channel accuracies across seeds are reported in Figure 4b of the main paper and are similar in trend to those of the discrete channel (Figure 1). Further examples of sketches produced at test time for each of the numerosity classes present in the training set are shown in Figure 5. The different rows indicate results from agents being trained with more or less diverse training samples, some classes having more or fewer instances. The pairwise dissimilarity of sketches in the three varying frequency setups is shown in Figure 3.

The results indicate the messages are arbitrary and there is no concrete correlation between message complexity and the number of training samples. The only consistent trend is observed in the accuracy across the different training setups, *Decreasing* being the most difficult to solve with both a discrete and continuous channel.

An observation from this experiment is that while the symbols do not vary in complexity based on the frequency, nor are they compositional (in the sense of combining more frequent numerosities' codes), they all seem to correlate with the area filled by elements. For example, numerosity 1 is usually represented through

a symbol whose lines are concentrated at the centre of the sketch and do not cover much of the available space. As the numerosity increases and the elements start spreading across the target image space, the lines start increasing in length and are positioned in such a way as to cover as much of the available sketching space.

B. Extended Results

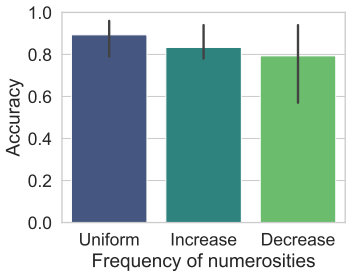
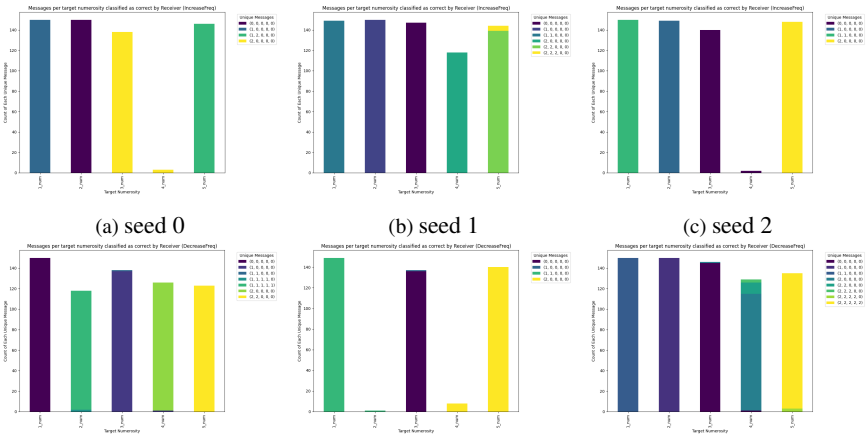


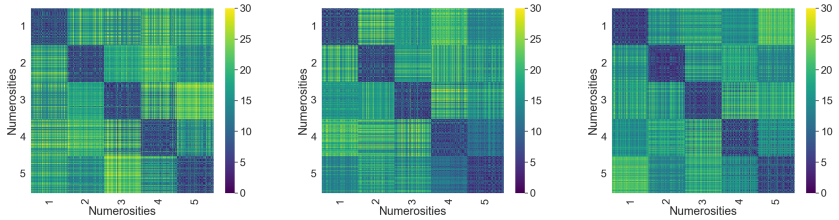
Figure 1. Accuracy and standard deviation across 3 seeds for communication task performed with different frequencies of training numerosity samples: *Uniform*, *Increase* and *Decrease* frequency setups.



(a) seed 0 (b) seed 1 (c) seed 2

(d) seed 0 (e) seed 1 (f) seed 2

Figure 2. Exact message to numerosity mappings for correctly classified numerosities, across 3 different seeds, in the *Increase* (a, b and c) and *Decrease* frequency setups (d, e, f).

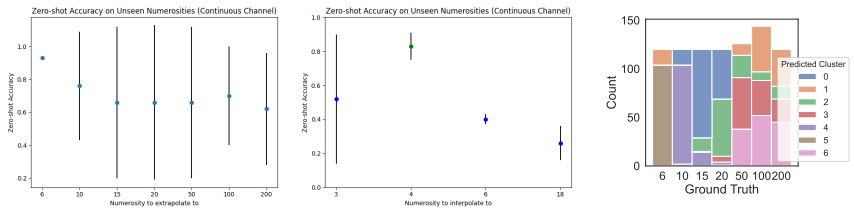


(a) Uniform (b) Increase (c) Decrease

Figure 3. Pair-wise dissimilarity for sketches in the tSNE-reduced pixel-space for various frequencies of numerosities: (a) equal, (b) increasing from 1 to 5, (c) decreasing from 1 to 5.

Table 1. Model with fixed message length trained on numerosities $\{1, 2, 3, 4, 5\}$ with different $vocabulary$ sizes: $\{3, 5, 10, 100\}$. Test communication accuracy and entropy are across 3 seeds.

Setup	**Max Msg Length**	**Vocabulary Size**	**Accuracy**	**Entropy**
Diff target	5	3	.95±.2	1.14±.34
	5	5	.96±0	1.15±.31
	5	10	.94±.02	1.13±.42
	5	100	.96±.008	0.76±.15



(a) Extrapolation (b) Interpolation (c) Mapping of sketches

Figure 4. (a, b) Zero-shot accuracies for out-of-distribution testing data for the continuous (*i.e.* sketching) channel. (a) Extrapolation: model trained on *train*=$\{1, 2, 3, 4, 5\}$ and tested on *train* + one OOD. (b) Interpolation: the different colours in interpolation (green vs blue) correspond to different training data - please refer to Fig. 5 in the main paper. c) Mapping of sketches after clustering based on similarity to extrapolated numerosities (cond. entropy: 1.348).

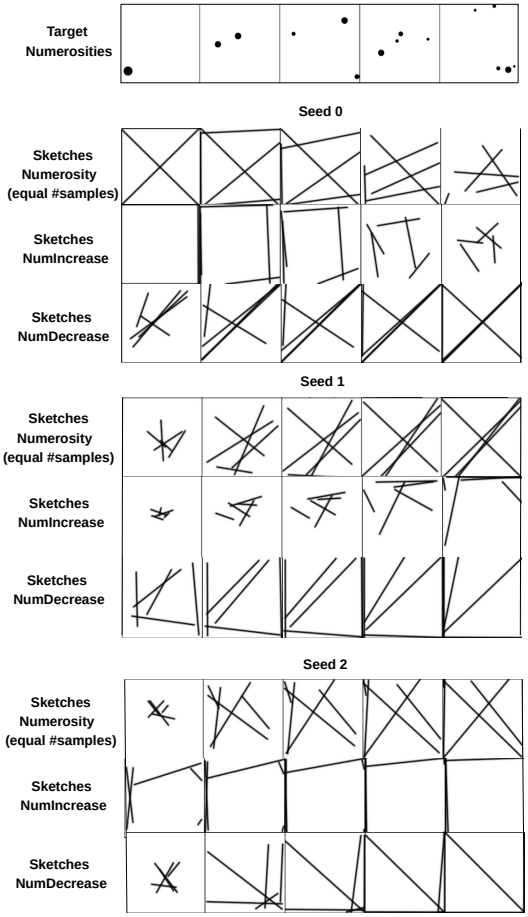


Figure 5. Test target images (each entry in the first row corresponds to a distinct numerosity between 1 and 5) and sketches produced by agents trained on different numerosity datasets, with varying numbers of class samples. Results are reported across 3 training seeds.